

IoT-Based Air Quality Monitoring System Using ESP8266

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List of Principal Symbols and Acronyms

- **IoT**: Internet of Things
- **AQI**: Air Quality Index
- **CPCB**: Central Pollution Control Board
- **PPM**: Parts Per Million
- **ADC**: Analog-to-Digital Converter
- **GPIO**: General Purpose Input/Output
- **I2C**: Inter-Integrated Circuit
- **SDA**: Serial Data Line
- **SCL**: Serial Clock Line
- **CO₂**: Carbon Dioxide
- **NH₃**: Ammonia

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Abstract

Air quality degradation presents severe threats to human health and environmental safety. Traditional environmental monitoring stations are capital-intensive, stationary, and lack real-time local public alerting systems. This project presents a cost-effective, real-time Internet of Things (IoT)-based Air Quality Monitoring System built on the ESP8266 NodeMCU platform. The device incorporates an MQ-135 semiconductor gas sensor to measure atmospheric pollutants and a DHT11 sensor to monitor temperature and relative humidity. A 16x2 I2C LCD provides immediate local data display, while a dual-LED (Green/Red) and active buzzer configuration delivers physical hazard warnings

The raw analog outputs from the MQ-135 sensor are calibrated against the standard Central Pollution Control Board (CPCB) Air Quality Index (AQI) categories (0 – 500 scale). Under safe air conditions ($AQI \leq 200$), the Green LED remains illuminated while the Red LED and acoustic buzzer stay off. When gas concentrations exceed safety thresholds ($AQI > 200$), the system updates the status classification on the LCD screen, activates the Red LED and warning buzzer, and transmits real-time telemetry data to the Blynk cloud dashboard. Experimental validation demonstrates rapid response times and high reliability for indoor ambient monitoring.

Chapter 1: INTRODUCTION

1.1 Introduction

Air pollution caused by vehicular exhaust, industrial fumes, and improper ventilation directly harms human health. Continuous monitoring of toxic gases (CO₂, Ammonia, Smoke, Benzene) is critical for maintaining safe indoor and outdoor environments. The Internet of Things (IoT) enables low-cost embedded hardware to collect, process, and broadcast environmental metrics remotely without complex infrastructure.

1.2 Review of Previous Works (Literature Review)

1. **Wireless Sensor Networks (WSNs):** Earlier environmental nodes relied heavily on GSM or ZigBee protocols, which increased operational costs and required specialized receiver equipment.
2. **Wi-Fi Embedded Systems:** Modern system-on-chip solutions like the ESP8266 NodeMCU integrate high-speed processing with native Wi-Fi, allowing seamless cloud telemetry.
3. **Local Visual Alerting:** Studies indicate that while cloud dashboards provide historical trend tracking, local physical alerts (LCD displays, LEDs, and buzzers) are essential for immediate safety intervention during local gas spikes.

1.3 Objective and Motivation

- **Objective:** To construct a standalone, IoT-enabled air monitoring node that displays live air quality, temperature, and humidity on a local LCD screen, provides immediate visual/audible alarms during pollution events, and syncs data to an IoT cloud dashboard.
- **Motivation:** Providing an accessible, affordable warning system for homes, laboratories, and classrooms to prevent long-term exposure to toxic airborne pollutants.

1.4 Scope of Present Work

This project covers circuit layout design, sensor interfacing, micro-controller programming using C++, local LCD menu design, hardware alert state logic, and cloud IoT dashboard configuration.

Chapter 2: HARDWARE ARCHITECTURE AND COMPONENT DESCRIPTION

2.1 ESP8266 NodeMCU Microcontroller

The ESP8266 NodeMCU is a 32-bit Wi-Fi module operating at 80 MHz. It features 16 GPIO pins, software I2C support, and a 10-bit Analog-to-Digital Converter (ADC) pin (A0).

2.2 MQ-135 Gas Sensor

The MQ-135 uses a Tin Dioxide (SnO_2) sensitive layer that alters its electrical resistance in the presence of polluting gases. It requires a 5V power line to supply its internal heating element.

2.3 DHT11 Temperature and Humidity Sensor

A basic digital sensor that outputs calibrated temperature ($^{\circ}\text{C}$) and humidity (%) data over a single digital line.

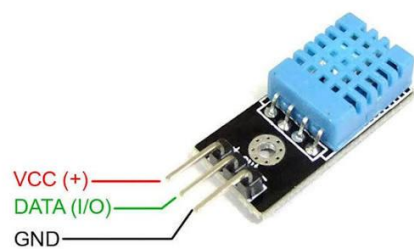


Fig 2.3

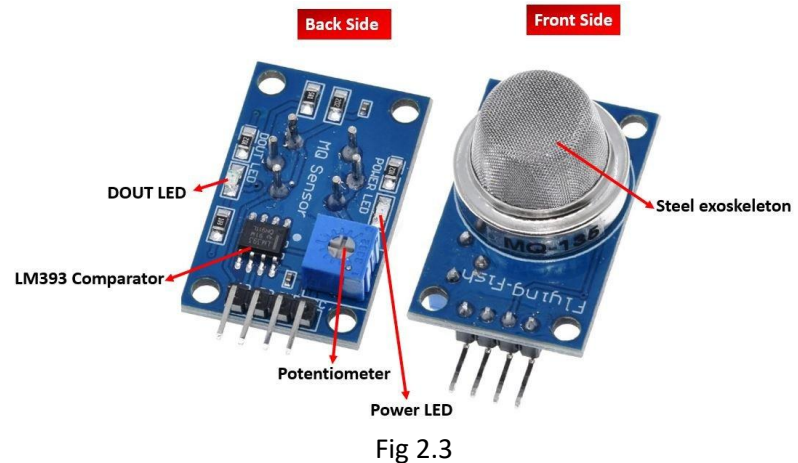


Fig 2.3

2.4 16x2 I2C LCD Display Module

A standard 16-character by 2-line liquid crystal display fitted with a PCF8574 I2C adapter. This adapter reduces the required microcontroller pins from 16 down to just 2 communication lines (SDA and SCL).

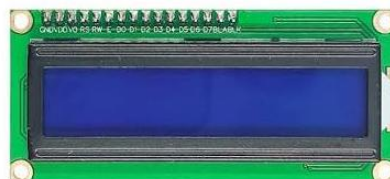


Fig 2.4

2.5 Alert Indicators (LEDs & Active Buzzer)

- **Green LED:** Indicates clean/safe air conditions ($A_0 \leq \text{Threshold}$).
- **Red LED:** Indicates elevated gas levels ($A_0 > \text{Threshold}$).
- **Active Buzzer:** Produces an audible alarm signal during hazard states.



Fig 2.5

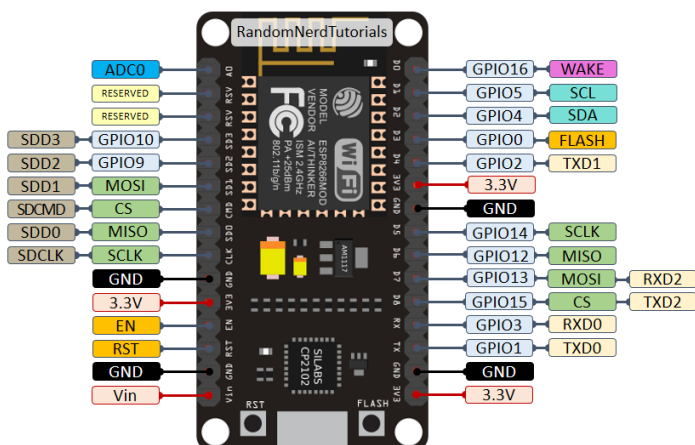


Fig 2.1

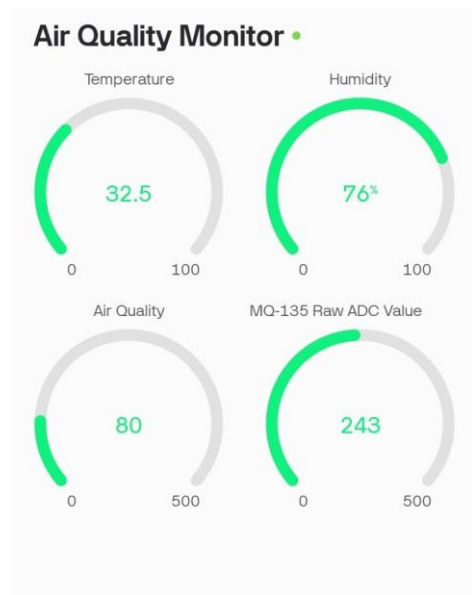


Fig 4.1

Chapter 3: CIRCUIT DESIGN AND SOFTWARE IMPLEMENTATION

3.1 Interfacing Circuit and Pinout

Table 3.1: Complete Circuit Pin Interfacing Mapping Table

Module / Component	Pin Label	NodeMCU ESP8266 Pin	Operating Voltage / Description
MQ-135 Gas Sensor	VCC	V _{in}	5V Power (Direct from USB)
	GND	GND	Common Ground Reference
	AO	A0	Analog Input (0 to 1023)
DHT11 Sensor	VCC	3V3 / V _{in}	3.3V/5V Power Line
	GND	GND	Common Ground Reference
	DATA	D4 (GPIO 2)	Digital Data Signal
16x2 LCD (with I2C)	VCC	V _{in}	5V Power (For Backlight)
	GND	GND	Common Ground Reference
	SDA	D2 (GPIO 4)	I2C Serial Data Line
	SCL	D1 (GPIO 5)	I2C Serial Clock Line

Module / Component	Pin Label	NodeMCU ESP8266 Pin	Operating Voltage / Description
Green LED	Anode (+)	D6 (GPIO 12)	Safe Indicator (Active-Low Control)
	Cathode (-)	GND	Common Ground Reference
Red LED	Anode (+)	D5 (GPIO 14)	Hazard Indicator (Active-Low Control)
	Cathode (-)	GND	Common Ground Reference
Active Buzzer	Positive (+)	D7 (GPIO 13)	Acoustic Alarm Output (Active-High/Low Configurable)
	Negative (-)	GND	Common Ground Reference

3.2 Software Environment & Cloud Integration

- Development IDE: Arduino IDE compiled using the ESP8266 Core library.
- Cloud Platform: Blynk IoT application configured with virtual streams:
 - **V0:** Calibrate Air Quality Index (AQI Value)
 - **V1:** Raw MQ-135 Voltage Count (ADC Value)
 - **V2:** Temperature (°C)
 - **V3:** Relative Humidity (%)

3.3 Firmware Source Code

```
#define BLYNK_TEMPLATE_ID "Your_Template_ID"
```

```

#define BLYNK_TEMPLATE_NAME "Air Quality Monitoring System"
#define BLYNK_AUTH_TOKEN  "Your_Auth_Token"

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

// WiFi Credentials
char auth[] = BLYNK_AUTH_TOKEN;
char ssid[] = "Your_WiFi_Name";
char pass[] = "Your_WiFi_Password";

// Hardware Pin Configuration
#define SENSOR_PIN A0  // MQ-135 Gas / Air Quality Sensor
#define DHTPIN  D4  // DHT11 Sensor Pin
#define DHTTYPE  DHT11 // Sensor Type
#define RED_LED  D5  // Air Quality Hazard LED
#define GREEN_LED D6  // Safe / Normal Air LED
#define BUZZER  D7  // Piezo Alert Alarm

// Explicit Hardware Logic Control Definitions
#define LED_ON  LOW  // Active-Low LEDs
#define LED_OFF HIGH

#define BUZZER_ON HIGH // Active-High Buzzer Output
#define BUZZER_OFF LOW

// Sensor Calibration Constants

```

```

const int AIR_BASELINE_ADC = 150; // Room baseline boundary (~198-200 ADC)
const int AIR_MAX_ADC      = 800; // Upper pollution limit

DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);
BlynkTimer timer;

void processAirQualityData() {
  // 1. Read Raw MQ-135 Sensor Data
  int rawADC = analogRead(SENSOR_PIN);

  // 2. Calibrate Raw ADC to 0-500 CPCB AQI Scale
  int calculatedAQI = map(rawADC, AIR_BASELINE_ADC, AIR_MAX_ADC, 10, 500);
  if (calculatedAQI < 0) calculatedAQI = 0;
  if (calculatedAQI > 500) calculatedAQI = 500;

  // 3. Read Climate Metrics from DHT11
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();

  // 4. Stream Telemetry to Blynk Cloud Dashboard
  Blynk.virtualWrite(V0, calculatedAQI); // Calibrated AQI
  Blynk.virtualWrite(V1, rawADC);       // Raw Voltage Count
  Blynk.virtualWrite(V2, temperature);  // Temperature (°C)
  Blynk.virtualWrite(V3, humidity);     // Humidity (%)

  // 5. LCD Row 0: Ultra-Compact Formatting (Fits 16 Characters Exactly)
  lcd.setCursor(0, 0);
  lcd.print("A:");
  lcd.print(calculatedAQI);

```

```
if (calculatedAQI < 10) lcd.print(" ");  
else if (calculatedAQI < 100) lcd.print(" ");
```

```
if (!isnan(temperature)) {  
    lcd.print("T:");  
    lcd.print((int)temperature);  
    lcd.print("C ");  
}
```

```
if (!isnan(humidity)) {  
    lcd.print("H:");  
    lcd.print((int)humidity);  
    lcd.print("%");  
}
```

```
// 6. LCD Row 1: CPCB Classification & Hardware Alert Logic  
lcd.setCursor(0, 1);
```

```
if (calculatedAQI <= 50) {  
    // BAND 1: Good  
    lcd.print("STATUS: GOOD ");  
    digitalWrite(GREEN_LED, LED_ON);  
    digitalWrite(RED_LED, LED_OFF);  
    digitalWrite(BUZZER, BUZZER_OFF);  
}
```

```
else if (calculatedAQI <= 100) {  
    // BAND 2: Satisfactory  
    lcd.print("STATUS: SATISFY ");  
    digitalWrite(GREEN_LED, LED_ON);  
    digitalWrite(RED_LED, LED_OFF);
```



```

    digitalWrite(BUZZER, BUZZER_OFF);
}
else if (calculatedAQI <= 200) {
    // BAND 3: Moderate
    lcd.print("STATUS: MODERATE");
    digitalWrite(GREEN_LED, LED_ON);
    digitalWrite(RED_LED, LED_OFF);
    digitalWrite(BUZZER, BUZZER_OFF);
}
else if (calculatedAQI <= 300) {
    // BAND 4: Poor
    lcd.print("STATUS: POOR  ");
    digitalWrite(GREEN_LED, LED_OFF);
    digitalWrite(RED_LED, LED_ON);
    digitalWrite(BUZZER, BUZZER_OFF);
}
else if (calculatedAQI <= 400) {
    // BAND 5: Very Poor
    lcd.print("STATUS: V. POOR ");
    digitalWrite(GREEN_LED, LED_OFF);
    digitalWrite(RED_LED, LED_ON);

    // Warning Beep
    digitalWrite(BUZZER, BUZZER_ON);
    delay(100);
    digitalWrite(BUZZER, BUZZER_OFF);
}
else {
    // BAND 6: Severe
    lcd.print("STATUS: SEVERE! ");

```

```

    digitalWrite(GREEN_LED, LED_OFF);
    digitalWrite(RED_LED, LED_ON);
    digitalWrite(BUZZER, BUZZER_ON); // Continuous Alarm
}
}

```

```

void setup() {
    Serial.begin(115200);

    // Pin Initialization
    pinMode(RED_LED, OUTPUT);
    pinMode(GREEN_LED, OUTPUT);
    pinMode(BUZZER, OUTPUT);

    // Default Initial Hardware State
    digitalWrite(RED_LED, LED_OFF);
    digitalWrite(GREEN_LED, LED_OFF);
    digitalWrite(BUZZER, BUZZER_OFF);

    // Initialize Sensors & Displays
    dht.begin();
    Wire.begin(D2, D1); // SDA = D2, SCL = D1
    lcd.init();
    lcd.backlight();

    lcd.setCursor(0, 0);
    lcd.print(" AIR QUALITY ");
    lcd.setCursor(0, 1);
    lcd.print(" MONITOR SYSTEM ");
    delay(2000);
}

```

```
lcd.clear();

// Connect to Blynk Cloud
Blynk.begin(auth, ssid, pass);

// Read sensors every 1.5 seconds
timer.setInterval(1500L, processAirQualityData);
}

void loop() {
  Blynk.run();
  timer.run();
}
```

Chapter 4: RESULTS AND DATA ANALYSIS

4.1 Sensor Calibration and CPCB Threshold Mapping

The MQ-135 semiconductor sensor undergoes a pre-heating period to stabilize its baseline resistance (R_S) in clean ambient air. To provide standard air monitoring compliance, the system maps the raw 10-bit analog voltage reading (0 – 1023 count) from pin A0 into the official Central Pollution Control Board (CPCB) Air Quality Index (AQI) categories (0 – 500 scale).

Table 4.1: Central Pollution Control Board's Air Quality Standards

Air Quality Index (AQI)	Category
0 - 50	Good
51 - 100	Satisfactory
101 - 200	Moderate
201 - 300	Poor
301 - 400	Very Poor
401 - 500	Severe

Table 4.2: Sensor Calibration, CPCB AQI Mapping, and Hardware Output Matrix

Raw ADC Range (A0)	Calibrated AQI Scale	CPCB Category	LCD Display Text	Green LED	Red LED	Buzzer Alarm State
150 – 200	0 – 50	Good	STATUS: GOOD	ON	OFF	OFF
201 – 265	51 – 100	Satisfactory	STATUS: SATISFY	ON	OFF	OFF
266 – 395	101 – 200	Moderate	STATUS: MODERATE	ON	OFF	OFF
396 – 525	201 – 300	Poor	STATUS: POOR	OFF	ON	OFF
526 – 655	301 – 400	Very Poor	STATUS: V. POOR	OFF	ON	FAST BEEP
656 – 800+	401 – 500	Severe	STATUS: SEVERE!	OFF	ON	CONTINUOUS

4.3 Real-time Testing and Hardware Response

- Normal Clean Air Conditions: In ambient room conditions, the raw analog reading on A0 averages between 190 – 220 ADC counts. This maps to a calibrated AQI value between 25 – 45 (Good/Satisfactory). The top line of the LCD displays A:26 T:27C H:65%, the status line displays STATUS: GOOD or STATUS: SATISFY, and the Green LED stays illuminated while the buzzer remains silent.

- **Pollutant Exposure Response:** When targeted pollutants or smoke are introduced near the MQ-135 sensor element, the analog output spikes rapidly above 400 ADC counts within 1.5 seconds.
- **Alert Execution:** As soon as the calculated AQI exceeds 200, the system instantly switches off the Green LED and turns on the Red LED. When AQI crosses into the **Very Poor** or **Severe** bands (> 300), the active buzzer triggers its acoustic alarm, the status line updates to STATUS: V. POOR or STATUS: SEVERE!, and live metric updates stream to the Blynk IoT app.

Chapter 5: FUTURE SCOPE AND CONCLUSIONS

5.1 Conclusions

The designed IoT Air Quality Monitoring System successfully achieves simultaneous local alerting and remote cloud telemetry. Incorporating an I2C LCD, dual LEDs, and an active buzzer guarantees immediate hazard warnings on site even during temporary Wi-Fi connectivity losses.

5.2 Future Scope

- **Relay-driven Exhaust:** Interfacing a 5V relay module to trigger an exhaust fan automatically whenever gas levels cross the hazard threshold.
- **Dust Particle Sensing:** Adding an optical dust sensor (e.g., GP2Y1014AU0F) to detect PM2.5 and PM10 particulate matter.

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